

Novel core-shell (TiO₂@Silica) nanoparticles for scattering medium in a random laser: higher efficiency, lower laser threshold and lower photodegradation†

Cite this: *Nanoscale*, 2013, 5, 12512

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Received 13th July 2013
Accepted 27th September 2013

DOI: 10.1039/c3nr03603k

www.rsc.org/nanoscale

There has been growing interest in scattering media in recent years, due to their potential applications as solar collectors, photocatalyzers, random lasers and other novel optical devices. Here, we have introduced a novel core-shell scattering medium for a random laser composed of TiO₂@Silica nanoparticles. Higher efficiency, lower laser threshold and long photobleaching lifetime in random lasers were demonstrated. This has introduced a new method or parameter (fraction of absorbed pumping), which opens a new avenue to characterize and study the scattering media. Optical chemical and colloidal stabilities were combined by coating a suitable silica shell onto TiO₂ nanoparticles.

Introduction

Strong scattering media (SSM) are of great current interest, due to their potential applications in photovoltaic cells,^{1,2} efficient photocatalyzers,³ light confinement,⁴ random lasers^{5,6} and novel optical functional devices.^{7,8} The coherent superpositions of input fields (SSM) could give rise to fundamental effects, such as light confinement,⁹ enhanced backscattering,¹⁰ coherently enhanced absorption,¹¹ lasing with coherent feedback¹² or even a transmission that is able to go much further through the sample than the scattering mean free path.¹³ Here, we present a random laser with high efficiency, a long photobleaching lifetime, narrower bandwidth and lower laser threshold than a traditional laser. It is composed of core-shell nanoparticles (TiO₂@Silica) suspended in an ethanol solution of Rhodamine 6G (R6G). The silica shell thickness was around 40 nm. The random or powder laser was predicted by Letokhov in 1968.¹⁴ In the work reported by Noginov,¹⁵ Cao¹⁶ and Wiersma,¹⁷ detailed reviews on random lasers can be found. Lawandy *et al.*, who suspended nanoparticles in a conventional laser dye, obtained the first evidence of a random laser in solution.¹⁸ The photodegradation effect and the inability to ensure complete colloidal dispersion, however, have limited the development and applications of such systems. TiO₂ nanoparticles are prone to aggregate in aqueous solution,

which limits extensive application. The addition of alcohol affects even more zeta potential values of particles, reducing the repulsive potential between particles that leads to suspension destabilization.¹⁹ A shell on strong scattering particles with a suitable thickness and refractive index would avoid the photodegradation effect and “optical” junction of scattering TiO₂ surfaces, preserving the medium scattering strength. Moreover, the silica shell should improve light coupling with the TiO₂ particles by light refraction at the ethanol-silica interface, being able to increase the medium scattering strength.

RL action in core-shell nanowires (ZnO/MgO) fixed on an ITO glass substrate has recently been reported.²⁰ However, although the MgO shell should improve the optical coupling between light and the nanowires, this system is not subject to agglomeration like a colloidal suspension. Particles like TiO₂@SiO₂ have previously been synthesized,²¹ and in the fluorescence study of the R6G molecules, which were absorbed onto the surface of the TiO₂@SiO₂ nanoparticles, they exhibited an increased lifetime and quantum efficiency, with silica thickness increasing from 1 nm to 10 nm.²² In this work, we propose coating the scattering particle surface (TiO₂) with a suitable silica shell (higher thickness ~40 nm) and using these particles (TiO₂@40 nm Silica) as a scattering medium in a random laser. To our knowledge, the core-shell strategy for a scattering medium (in colloidal suspension) in a random laser has not yet been reported. The silica shell would act as a barrier to prevent charge transfer, which has been reported as the cause of dye degradation.²³ In addition, coatings with thicknesses around or above 40 nm prevent the “optical” junction of the TiO₂ scattering surfaces and should improve the light coupling with the TiO₂ particles. In turn, the silica shell is particularly advantageous, due to its high solubility,

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† Electronic supplementary information (ESI) is available. See DOI: 10.1039/c3nr03603k

low density and the inertness of the nanoparticles, along with the numerous possibilities for their use.^{24–29} Therefore strong scattering particles coated with a shell of suitable thickness and refractive index could provide new opportunities and achieve significant improvements in the operation of random lasers and photonic devices based on highly disordered scattering media.

Materials and methods

Two kinds of samples were prepared, both containing $[1 \times 10^{-4} \text{ M}]$ of R6G, one with TiO_2 and the other with $\text{TiO}_2@\text{Silica}$ scattering nanoparticles (Np). The silica coating of TiO_2 Np was made *via* a variant of the Stöber method,^{30,31} using tetraethyl-orthosilicate (TEOS) as the silica precursor. The material description and procedure used to synthesize the core-shell $\text{TiO}_2@\text{Silica}$ nanoparticles are found in the ESI†. The silica coating on the TiO_2 nanoparticles was examined by transmission electron microscopy (TEM). The stoichiometric ratio (Ti/Si) of nanoparticles ($\text{TiO}_2@\text{Silica}$) was determined by energy dispersive X-ray fluorescence (ED-XRF). The sample preparation details for these characterizations can be found in the ESI† archive.

The pumping source of the random laser (RL) was the second harmonic of a Q-switched Nd:YAG Continuum Minilite II (25 mJ, 532 nm, with a pulse width of ~ 6 ns and a repetition rate up to 15 Hz), multimode, linearly polarized, with a spot size of 3 mm. The emission spectra were collected through a multi-mode optical fiber (200 μm), coupled to a spectrometer HR4000 UV-Vis (Ocean Optics) with a spectral resolution of 0.36 nm (FWHM). A schematic diagram of the experimental setup of RL is shown in Fig. S1 of the ESI†.

Results

Characterization

In the TEM images, we observe the silica coating on TiO_2 nanoparticles, such as the one indicated by the yellow arrows (Fig. 1a). This silica shell presents an irregular morphology, with a thickness ranging between 20 and 80 nm. Fig. 1b shows the surface of one TiO_2 nanoparticle before the silica coating was applied. It is evident that the nanoparticle surface is irregular; this should determine the subsequent morphology of the silica coating. The mass percentage ratio (Ti/Si) determined by ED-XRF was $\text{Ti}_{70}/\text{Si}_{30}$. Considering the typical density of the silica obtained by the TEOS hydrolysis³² between 2.0 and 2.2 g cm^{-3} , we obtain an average silica shell thickness between 42 and 38 nm. In this way, the silica coating thickness would exactly represent a barrier that prevents the “optical” binding of the TiO_2 scattering surfaces, with the additional advantage of providing a chemically stable surface.

Random laser action: low $\text{TiO}_2@\text{Silica}$ concentration

Fig. 1c and d show the behavior of the emitted intensity and the spectral width (FWHM), as a function of pumping energy fluence for the two kinds of scattering particle solutions (TiO_2 and $\text{TiO}_2@\text{Silica}$). The calculated concentration of scattering nanoparticles was $5.6 \times 10^{10} \text{ Np ml}^{-1}$. For this calculation, the nanoparticles were considered to be completely dispersed, without forming any kind of agglomerates. Fig. 1c inset shows a photograph of the suspensions of TiO_2 and $\text{TiO}_2@\text{Silica}$ nanoparticles in an ethanol solution of R6G. The solution containing TiO_2 nanoparticles presents a duller appearance and is more turbid than the $\text{TiO}_2@\text{Silica}$, in which the rhodamine pigment is more pronounced.

The pumping energy fluence varied between 0.12 and 120 mJ cm^{-2} for the study of random laser action. The random laser action (Fig. 1c and d) was improved for the system $\text{TiO}_2@\text{Silica}$, *i.e.* it presented a higher slope efficiency, narrower bandwidth and lower laser threshold. For the $\text{TiO}_2@\text{Silica}$ system, the RL efficiency slope (RL_{eff}) was 2.15 ± 0.06 fold greater than for TiO_2 . The laser threshold values extracted from the fits (Fig. 1d) for the TiO_2 and $\text{TiO}_2@\text{Silica}$ systems were $2.28 \pm 0.04 \text{ mJ cm}^{-2}$ and $1.81 \pm 0.03 \text{ mJ cm}^{-2}$, respectively. The specular reflection measurements for both samples showed no difference; therefore, the coupling of pumping radiation with the samples is similar (see ESI†). A comparison between peak wavelengths of emission spectra, as a function of fluence (between 0.12 and 120 mJ cm^{-2}), is found in the ESI (Fig. S2a–c†) for both systems (TiO_2 and $\text{TiO}_2@\text{Silica}$). The redshift of the RL spectrum with respect to the center of the luminescence spectrum at low pump intensity is almost null ($< 0.5 \text{ nm}$) for the $\text{TiO}_2@\text{Silica}$ system at 120 mJ cm^{-2} . Therefore, it is inferred that the percentage of R6G molecules involved in the stimulated emission is close to 100%. The above results could be associated with a lower scattering mean free path (l_s) (see Table 1) (higher scattering strength), caused by a higher effective scattering surface (higher “optical” colloidal stability). This means that the pumping energy is

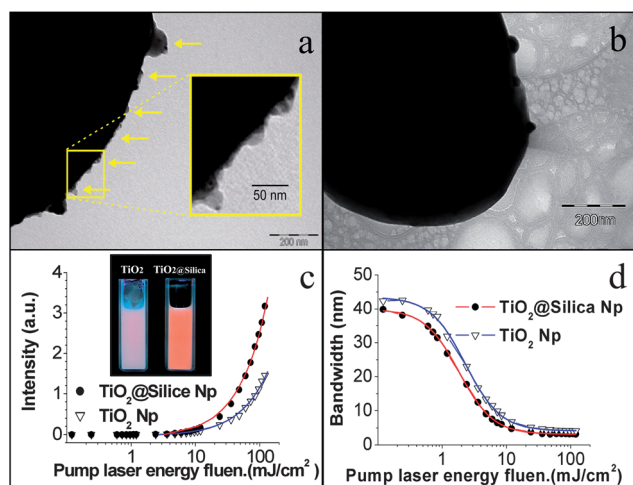


Fig. 1 (a) TEM images of silica coating on the $\text{TiO}_2@\text{Silica}$ surface and (b) the surface of the TiO_2 nanoparticle. The scale bars represent 200 nm. Yellow arrows (a) indicate the silica coating and the inset in (a) is a magnification of the particle surface, framed with the yellow square. (c and d) Influence of the pumping energy fluence: (c) the emitted peak intensity and (d) spectral FWHM emission of the RL, for the two kinds of nanoparticles (TiO_2 and $\text{TiO}_2@\text{Silica}$). Inset of (c): photograph of ethanol solutions of R6G containing nanoparticles of TiO_2 and $\text{TiO}_2@\text{Silica}$. The calculated particle density was $5.6 \times 10^{10} \text{ Np ml}^{-1}$ and $[1 \times 10^{-4} \text{ M}]$ of R6G. The solid lines represent the fitting with experiments points.

Table 1 FAP and ABS values for fluencies well below (FAP_{bT} , ABS_{bT}) and above the RL threshold (FAP_{aT} , ABS_{aT}); l_s values measured by the transmission experiment. The TiO_2 and $TiO_2@Silica$ Np concentration was 5.6×10^{10} Np ml^{-1}

TiO_2					$TiO_2@Silica$				
FAP_{bT}	FAP_{aT}	ABS_{bT}	ABS_{aT}	l_s (μm)	FAP_{bT}	FAP_{aT}	ABS_{bT}	ABS_{aT}	l_s (μm)
2.5	1.96	0.92	0.67	52 ± 4	4.7	4.2	1.55	1.44	20.6 ± 0.2

confined in a lower volume (higher effective pumping fluence). Moreover, the amount of R6G molecules inside this volume would be lower, being able to excite a higher percent of molecules (higher population inversion). The l_s measurement details and an expanded discussion about the colloidal stability are found in the ESI.† Additionally, we introduced a new method to understand the results more clearly. The light reflected from pulse pumping by the samples was measured, with and without R6G, for the two scattering media (TiO_2 and $TiO_2@Silica$). The ratio of the pumping intensities reflected by the scattering medium with and without R6G was designated as the fraction of absorbed pumping (FAP):

$$FAP = (I_{RNp}/I_{RNp+R6G}) = e^{(l_{eo}/l_a)} \Rightarrow (l_{eo}/l_a) = \ln[FAP],$$

where l_{eo} would be the average photon path length (pumping radiation) inside the scattering medium before being reflected and l_a is the ballistics absorption length (R6G). The ratio (l_{eo}/l_a) is known as absorbance (ABS). FAP measurements as a function of pumping fluence for TiO_2 and $TiO_2@Silica$ systems are shown in the ESI (Fig. S2d†). Table 1 shows the FAP and ABS values to fluencies well below (FAP_{bT} and ABS_{bT}) and above (FAP_{aT} and ABS_{aT}), the RL thresholds for both systems and the l_s values (5.6×10^{10} Np ml^{-1}). These values (FAP_{bT} ; FAP_{aT} and ABS_{bT} ; ABS_{aT}) are constant (see ESI†).

For both systems, the l_a well below the laser threshold (passive regime) should not change ($885 \mu m$ at $[10^{-4} M]$ of R6G). Therefore, from the FAP_{bT} value, l_{eo} could be determined. For the $TiO_2@Silica$ system, $l_{eo} = l_a \times \ln FAP_{bT} \approx 885 \mu m \times \ln(4.7) \approx 1370 \mu m$; however, the measured l_s value was $20.6 \pm 0.2 \mu m$, which is ~ 67 times lower. This could mean that the photons (pumping radiation) undergo, on average, ~ 67 scattering events before being reflected. In this way, the photon residence time (τ_{eo}) in the scattering medium can be expressed as:

$$\tau_{eo} = [l_{eo}/c] \times n_{eth}$$

$$\tau_{eo} = 1370 \mu m \times 3.3 \times 10^{-3} ps \mu m^{-1} \times 1.36 \approx 6.2 ps$$

where n_{eth} and c are the ethanol refractive index and speed of light, respectively.

On the other hand, the ratio of ABS_{aT} values ($ABS_{(TiO_2@Silica)}/ABS_{(TiO_2)}^{aT} = 1.44/0.67 = 2.15$) corresponds to the ratio of the RL efficiency slopes (2.15). Thereby, an important result is highlighted: the RL efficiency slope (RL_{eff}) is equal to the absorbance above the RL threshold ($RL_{eff} = ABS_{aT} = l_{eo}/l_{aT}$), where l_{aT} is the ballistics absorption length above the RL threshold.

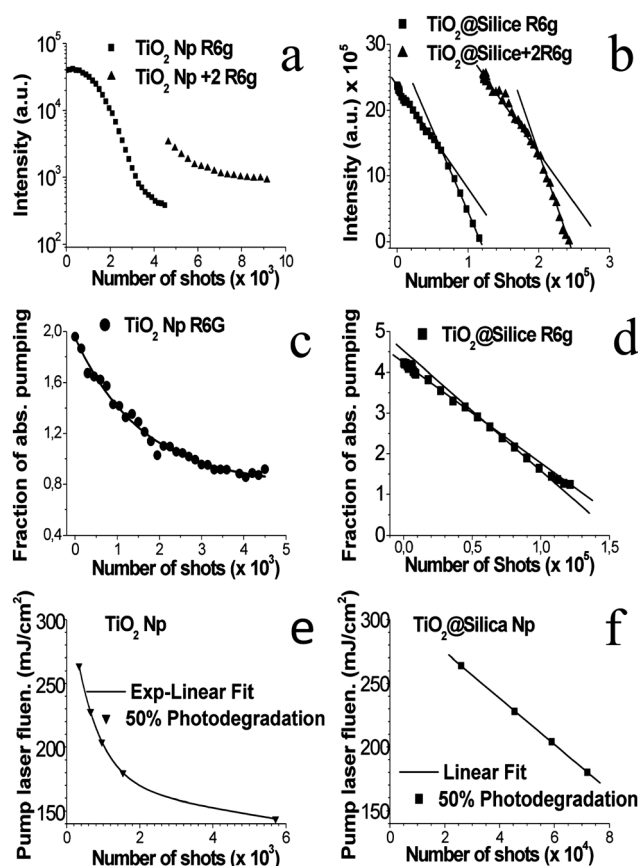


Fig. 2 (a and b) Photodegradation process of RL action as a function of the number of shots (laser pumping fluence $180 mJ cm^{-2}$): (a) ($\blacksquare$) TiO_2 Np + R6G system, ($\blacktriangle$) re-addition of R6G (TiO_2 Np + 2R6G); (b) ($\blacksquare$) $TiO_2@Silica$ Np + R6G system, ($\blacktriangle$) re-addition of R6G ($TiO_2@Silica$ Np + 2R6G). (c and d) FAP values as a function of the number of shots for: (c) the TiO_2 Np system and (d) the $TiO_2@Silica$ Np system. (e and f) Number of shots required for the 50% drop in RL emission intensity, as a function of the pumping fluence: (e) the TiO_2 Np system, (f) the $TiO_2@Silica$ Np system. The solid lines correspond to the functions fitted for each conjunct of experimental points.

Photodegradation studies

Fig. 2 shows the photodegradation process by the peak intensity of the random laser signal of R6G as a function of the number of shots for systems TiO_2 and $TiO_2@Silica$. The laser beam, of 3 mm diameter and fluence of $180 mJ cm^{-2}$, was used to pump the samples, whose volume was 200 μl accommodated in a 2 mm path length quartz cuvette. Fig. 2a and b show a decrease in emission intensity with the number of shots. The TiO_2 system shows a rapid exponential decay. The number of shots for which the emission intensity decreased to 50% was 1540. However, for

the TiO_2 @Silica system, the number of shots required was much higher ($\sim 72\,000$), which represents more than 46 times that of the TiO_2 system alone. The TiO_2 photocatalytic properties are a well-studied subject and have been used to remove or to degrade dyes from the environment.³³ The photocatalytic pathways involved in the reaction on the TiO_2 surface, as well as the possible explanations for our conditions, are found in the ESI.†

In order to study the photodegradation of the scattering medium, R6G was re-added to the samples after they were photo-degraded. In these graphs corresponding to the re-addition of R6G, it can be seen that the RL emission intensity for the TiO_2 nanoparticle system reaches only 10% of the initial value. However, for the TiO_2 @Silica system, the RL emission intensity returns to the initial value. This suggests that the TiO_2 @Silica nanoparticles, unlike the TiO_2 Np, do not suffer appreciable alterations in their optical properties under conditions that significantly alter TiO_2 alone. Fig. 2c and d show the FAP for the two systems, R6G + TiO_2 and R6G + TiO_2 @Silica, as a function of the number of shots, which should be proportional to the R6G concentration. The experimental FAP points for both, TiO_2 and TiO_2 @Silica, were fitted with an exponential function and two linear functions, respectively.

To more clearly understand the origin of the photo-degradation phenomenon, a study as a function of the pumping fluence was performed. Fig. 2e–f represent the number of shots necessary for the RL emission intensity drop of 50%, as a function of pumping fluence. The TiO_2 system shows an exponential dependence of the photodegradation (50%) for higher fluences ($>160\text{ mJ cm}^{-2}$) and linear dependence for lower fluences ($<140\text{ mJ cm}^{-2}$). However, the photodegradation for the system TiO_2 @Silica shows a linear behavior with pumping fluence. This indicates a quadratic relationship with the electric field, suggesting a bi-photonic process, which could be explained by the following hypothesis:

- through the reaction of the ethanol radical $\text{CH}_3\dot{\text{C}}\text{HOH}$ with R6G ground state molecules,³⁴ the free radical $\text{CH}_3\dot{\text{C}}\text{HOH}$ is produced by energy transfer from the R6G molecules in a higher triplet state, which is produced by two sequential single-photon absorptions.³⁵

Random laser action: high TiO_2 @Silica concentration

In order to study the TiO_2 @Silica scattering medium in depth, two suspensions with TiO_2 and TiO_2 @Silica nanoparticles, concentrated ~ 16 fold, were prepared. The previously synthesized TiO_2 @Silica nanoparticle suspension was introduced into a Rota-Evaporator to reduce the ethanol volume ~ 16 times. Fig. 3a and b show the behavior of the emitted intensity and the spectral width (FWHM), as a function of pump energy fluence for the two systems (TiO_2 and TiO_2 @Silica). The scatterers and dye concentrations were $\sim 8.9 \times 10^{11}$ nanoparticles per ml and $1 \times 10^{-4}\text{ M}$, respectively. The RL efficiency slope is not constant for the TiO_2 @Silica system; it decreases markedly to pumping fluences $>24\text{ mJ cm}^{-2}$. The green lines represent the slopes that should follow the RL emission for each system. Instead, the RL_{eff} is approximately constant for the TiO_2 system, showing a small increase at fluences $>60\text{ mJ cm}^{-2}$.

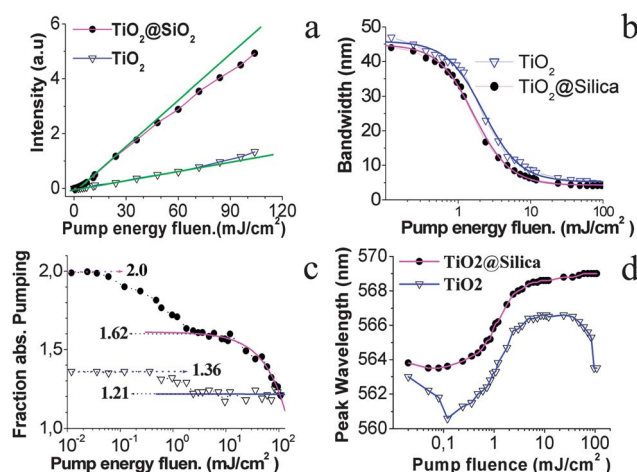


Fig. 3 Influence of the pump energy fluence at a high nanoparticle concentration ($\sim 8.9 \times 10^{11}\text{ Np cm}^{-3}$) and R6G [10^{-4} M] on: (a) the emitted peak intensity, (b) spectral FWHM emission of the RL, (c) fraction of absorbed pumping (FAP) (FAP values well below FAP_{bT} and above the FAP_{aT} RL threshold are embodied in the graphic) and (d) peak position of the emission spectrum. The solid lines: the blue line and the magenta line correspond to the TiO_2 and TiO_2 @Silica systems, respectively. The close circles and open triangles represent the TiO_2 @Silica and TiO_2 systems, respectively.

The calculated RL threshold for the TiO_2 @Silica system ($1.65 \pm 0.03\text{ mJ cm}^{-2}$) is lower than that for the TiO_2 system ($2.23 \pm 0.08\text{ mJ cm}^{-2}$).

The FAP dependence with pumping energy fluence is shown in Fig. 3c. For fluences well below the laser threshold, the FAP_{bT} values for both the TiO_2 and the TiO_2 @Silica systems are constants. For the TiO_2 system, the FAP value decreases for fluences from 0.36 mJ cm^{-2} up to 2.4 mJ cm^{-2} . However, for TiO_2 @Silica, the FAP value starts to drop for fluences that are much lower, from 0.036 mJ cm^{-2} up to 1.8 mJ cm^{-2} . This fact could suggest population inversion for very low fluences ($\geq 0.036\text{ mJ cm}^{-2}$). For fluences between 1.8 and 12 mJ cm^{-2} (above the laser threshold), FAP_{aT} values (TiO_2 @Silica) are approximately constant (1.62). For fluences $\geq 24\text{ mJ cm}^{-2}$, FAP_{aT} values show a drastic drop, corresponding with decreasing laser efficiency. In contrast, the FAP_{aT} for the TiO_2 system is approximately constant over the energy pumping range (1.21). For this Np concentration, the ABS_{aT} value ratio ($\text{ABS}_{\text{TiO}_2\text{@Silica}}/\text{ABS}_{\text{TiO}_2}$) $^{\text{aT}} \approx 2.53$ is higher than that for a low Np concentration (2.15); however, it does not correspond to the RL_{eff} ratio (5.1 ± 0.2). It could be explained by the decrease in fluorescence lifetime and the quantum efficiency of R6G molecules close to the TiO_2 surface for this high Np concentration²² (see ESI†). For the TiO_2 @Silica system, $l_{\text{eO}} = 885\text{ }\mu\text{m} \times \ln(2) \approx 610\text{ }\mu\text{m}$ and consequently, $\tau_{\text{eO}} = (610\text{ }\mu\text{m}/c) \times 1.36 \approx 2.77\text{ ps}$.

The measured l_{s} value was $2.1 \pm 0.04\text{ }\mu\text{m}$, which is ~ 292 times lower than l_{eO} , i.e. the average number of scattering events (N_{d}) per unitary volume (l_{s}^3) of each N_{i} incident channel would be ~ 292 . Therefore, at this high concentration, (TiO_2 @Silica) N_{d} is greater than the density of propagating channels: ($N_{\text{p}} \approx A_{\text{is}}/\lambda^2$; $A_{\text{is}}^2 = l_{\text{s}} \Rightarrow N_{\text{p}} \approx 16\text{ channels} \ll N_{\text{d}} \approx 292$).

In this way, the likelihood of interferential events should increase, which could lead to phenomena that involve coherent events, such as feedback lasing or enhanced absorption.

Instead, for the TiO_2 system, $l_s = 15 \pm 1 \mu\text{m}$, $l_{\text{eO}} \approx 272 \mu\text{m}$ and $N_p \approx 795 \gg N_d \approx 18$, which means that interferential events are highly improbable.

The peak position of the RL emission spectrum was studied as a function of fluence (Fig. 3d) for two systems (TiO_2 @Silica and TiO_2). The RL emission peak for TiO_2 @Silica shows a redshift that increases rapidly with the fluence until saturated. The saturated redshift value (5.4 nm) was really surprising, especially considering that the absorption and emission are saturated. The saturated absorption would mean a very low percentage of R6G molecules in the ground state, S_0 , and/or a high population of S_1 excited singlet states. Saturated emission means a very low population of the first S_1 excited singlet. In this way, the great redshift (5.4 nm) of the RL emission peak could mean that the few R6G molecules in the ground state, S_0 , exhibit an absorption band equivalent to a higher R6G concentration. The redshift of the RL emission peak for the TiO_2 system shows a maximum value of 2.8 nm, which is less than with the lower concentration (3.4 nm). This indicates that the percent of excited R6G molecules is higher. The redshift decrease for high fluencies ($\geq 48 \text{ mJ cm}^{-2}$) could be associated with decreasing l_s , which would provoke the pump energy to be confined in a smaller volume (see ESI†).

Conclusions

The novel core-shell nanoparticles (TiO_2 @Silica) for a random laser combine the high refractive index of TiO_2 with chemical inertness, "optical" colloidal stability (steric "optical" effect), and better light coupling (TiO_2 cores) provided by the silica shell. According to our knowledge, it is the first time that this strategy has been introduced in a scattering medium in colloidal suspension for a random laser. Higher efficiency, lower RL threshold and lower photodegradation in the random laser were achieved. It is found that the RL efficiency slope is equal to the sample absorbance above the laser threshold ($\text{RL}_{\text{eff}} = \text{ABS}_{\text{AT}}$).

The FAP measurement and its relation with l_s could be a new method to characterize scattering media and estimate the likelihood of interferential events. Additionally, a new phenomenon of absorption-emission saturation in the random laser was observed. A detailed study of RL action from micrometer size regions is called for, in order to discern possible enhanced absorption and/or lasing with coherent feedback. This core-shell strategy could be extrapolated to other scattering media composed of particles, which would allow it to obtain a higher scattering surface per unit volume, *i.e.*, higher scattering strength. The practical solutions presented and discussed in this paper could enable the design, manufacture and characterization of novel optical devices, based on highly disordered scattering media.

Acknowledgements

We gratefully acknowledge financial support from Rede 36 Nanobiotec CAPES (Brazil). V.M. thanks the CAPES (Brazil) for

doctoral fellowships. Thanks to Professors Luis Poveda and Wagner Faustino for their thoughtful suggestions. Thanks to Engineer Ricardo Acosta, for his corrections to the text.

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